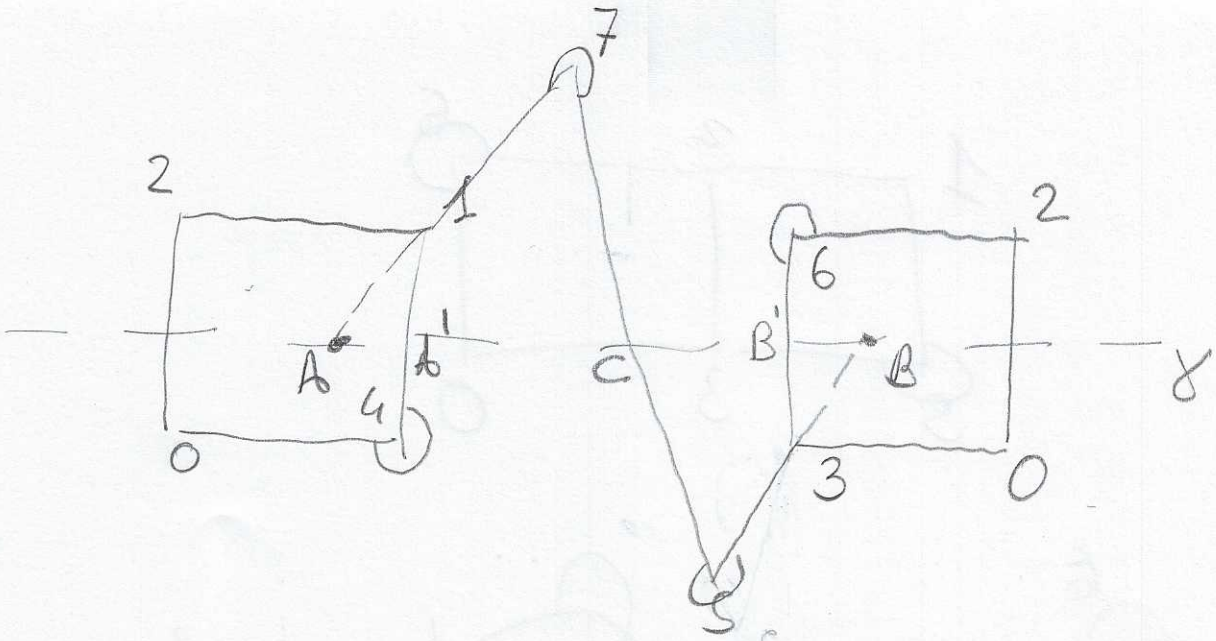
Esercizio rete 5313



The midpoints of the edges 20, 14, 63 are all on the same equator γ
We also claim that 5 and 7 are at opposite latitude: it follows because:

a) 3BB' and 1A'A are congruent because the angles $1\hat{A}'A$ and $B\hat{B}'3$ are equal ($\pi/2$) $A\hat{1}A'$ and $B\hat{3}B'$ are equal ($\pi/3$) and the segments 1A' and B'3 are congruent. So $3\hat{B}B' = 1\hat{A}A'$

b) Hence the triangles 7AC and 5BC are congruent:

$7\hat{A}C = 5\hat{B}C \quad C\hat{5}3 = C\hat{7}A = \frac{2\pi}{3}$

and $7\hat{C}A = 5\hat{C}B$

It follows as well that $CA' = CB'$.
The entire picture is thus symmetric with respect to the point C.
In particular, because the two pins are determined by the intersections of the bisectors of 7 and 6 and by those of 5 and 4 lying in the respective half-spheres, those pins are also mapped into each other by the symmetry with respect to C. P1 and P2

Since P_1 and P_2 are antipodal, they must be the two poles (with respect to the equator γ) So the line connecting N to 6 is part of a meridian and in particular if we continue it past 6 it meets γ orthogonally. But this is not possible

because the segment $6B'$ is orthogonal to γ, so B' would need to be on the same meridian as N and 6, on the other hand NS forms a $120°$ degree angle with $6B'$.